

The Korea Aerospace Research Institute (KARI; Korean: 한국항공우주연구원), established in 1989, is the aeronautics and space agency of South Korea. Its main laboratories are located in Daejeon, in the Daedeok Science Town. KARI's vision is to continue building upon indigenous launch capabilities, strengthen national safety and public service, industrialize satellite information and applications technology, explore the Moon, and develop environmentally-friendly and highly-efficient cutting-edge aircraft and core aerospace technology. Current projects include the KSLV-2 launcher. Past projects include the 1999 Arirang-1 satellite. The agency was founded in 1989. Prior to South Korea's entry into the Institute for Advanced Engineering (IAE) in 1992, it focused primarily on aerospace technology. As of May 2024, KARI is an affiliated research institute of the Korea AeroSpace Administration.

Background

KARI began on October 10, 1989, as a national aerospace research institute with the purpose of contributing to sound development of the national economy and enhancement of people's lives through a new exploration, technological advancement, development, and dissemination in the field of aerospace science and technology. It started its research in a temporary building, and had about 30 employees. In 1992, it was designated as a performance quality inspection specialized agency under the Aerospace Industry Development Promotion Act of the Ministry of Commerce, Industry and Energy, and in November 1996, it became an independent foundation called the Korea Aerospace Research Institute.

In 2001, it was renamed to the current Korea Aerospace Research Institute. In 2008, it was incorporated under the Basic Technology Research Council of the Ministry of Education, Science and Technology. In 2014, its affiliation was changed to the National Research Council of Science and Technology. In December 2016, it was selected as a national space

development specialized agency.

Organization

The headquarters is located in Daedeok Innopolis in Daejeon, and has various aerospace-related test facilities, including subsonic wind tunnel test facilities, full-body structural test facilities, propulsion test facilities, aircraft control test facilities, and rotary wing test facilities.

Naro Space Center

The Naro Space Center in Goheung, South Jeolla Province, has major rocket propulsion system test facilities, a launch pad, a tracking station, and launch operation facilities, and a launch pad for private space companies is also planned to be built within the center.

Goheung Aviation Center

The center is equipped with a 0.7 km long runway, airships, small and unmanned aircraft test facilities, landing gear drop test facilities, and propeller test facilities. From 2003 to 2019, the Goheung Aviation Center conducted performance tests on 168 manned and unmanned aircraft over a period of 16 years, and 76,077 researchers visited the center.

Tracking station

The Jeju Tracking Station is located in Seogwipo, Jeju Province, 200km away from the Naro Space Center, and is responsible for continuous tracking and information reception of space launch vehicles.

The Palau Tracking Station is equipped with a 7.3m diameter remote data receiving antenna and satellite communication network on a 28,000 m² site. It can receive remote data and images such as real-time flight position, speed, attitude, stage separation, satellite separation

signal, pressure, temperature, voltage, and current of launch vehicles or payloads, and can also receive data from launch vehicles that are more than 3,000 km away from the Naro Space Center.

Deep Space Ground Station

The station communicates with a space probe, and is located in Yeosu, Gyeonggi Province. It communicates with the lunar orbiter that will orbit the moon and conduct exploration activities. The antenna reflector sends out radio waves and collects the radio waves sent by the lunar orbiter.

Developments

KARI is also developing Unmanned Aerial Vehicles, high-altitude airships, a next-generation multi-purpose helicopter project, next-generation medium satellite, and a lunar exploration project. In addition, several satellites, including the KOMPSAT (also known as Arirang) Series, the COMS (Communication, Ocean and Meteorological Satellite), and the STSAT (Science and Technology Satellite), are developed, operated, or under development by KARI. On January 30, 2013, they launched a satellite into space from their own soil for the first time.

KARI began in 1989 to develop its own rockets (KSR, Korean Sounding Rocket). It produced the KSR-I and KSR-II, one and two-stage rockets in the early 1990s.

In December 1997 it began development of a LOX/kerosene rocket engine. KARI wished to develop satellite launch capability. A test launch of the KSR-III took place in 2002.

The first-stage unguided rocket KSR-I was produced by KARI in cooperation with research institutes such as industry, academia, and research institutes for the development of rockets. R&D took place for three years from 1990, and the total development cost was 2.85 billion won. Their goal was to develop a solid unguided science tube survey rocket that could observe the ozone layer over the Korean Peninsula.

Earth observation program

KSLV-1 launched on August 25, 2009. It was originally a cluster of indigenous liquid KSR-III rockets with a solid propellant to form a multistage launcher. However, KARI experienced more difficulties than expected in SLV development, because it required much stronger propellant power than KSR-III possessed to launch a satellite into orbit. After several failed attempts, KARI turned to Russia's Khrunichev Design Bureau for assistance in developing a liquid-propellant rocket engine for the KSLV-1 as well as for cooperation in the construction of the launch facility at the Naro Space Center. The first stage of the launcher was derived from the Russian URM-1 (Universal Rocket Module) developed by Khrunichev. The second stage of the launcher was a solid Kick Motor developed by Korea, which included the Inertial Navigation System; the power, control, and flight safety systems; plus the nose fairing.

South Korea launched its next rocket, the KSLV-2 in 2021. The first stage of KSLV-II has 4 clustered engines, each of which has a 75 metric ton thrust. All three stages use indigenously developed rocket engines.

Active Missions: KSLV-1, KSLV-2, Koreasat, ANASIS-II

Satellite program

SaTReC

The KAIST's SaTReC, responsible for the nation's small scientific satellites.

Korean Positioning System (KPS)

South Korea plans to spend 4 trillion won (\$3.56 billion) on building its own Regional Navigation Satellite System (RNSS), named Korean Positioning System (KPS). It plans launching seven new satellites by 2035 — three into geosynchronous orbit and four into inclined geosynchronous orbit. South Korea and the U.S. signed an agreement on "civil global

navigation satellite systems cooperation", under which the U.S. will support South Korea developing KPS. The two governments plan to work together to ensure compatibility and enhance interoperability of GPS and KPS for civil purposes.

Lunar and interplanetary missions

The Korean Lunar Exploration Program (KLEP) is divided in two phases. Phase 1 incorporates the launch and operation of a lunar orbiter called Korea Pathfinder Lunar Orbiter (KPLO), which will be the first lunar probe by South Korea, meant to develop and enhance South Korea's technological capabilities, as well as map natural resources from orbit. In December 2016, KARI signed a lunar exploration technical cooperation with NASA which increased the possibility of exploration success greatly. Phase 2 will include a lunar orbiter, a lunar lander, and a rover to be launched together on a KSLV-II South Korean rocket from the Naro Space Center, by 2030.

KPLO will fly 100 kilometers above the moon to carry out a lunar observation mission. The project will be carried out by KARI in charge of the system, the main body of the orbiter, and the ground station, and South Korean universities and research institutes will cooperate together.

Aircraft

Stratospheric airship

A 10-year program to develop a stratospheric airship for telecommunication relays and ground observations at 20 km altitude has started in December 2000.

Smart Tilt Rotor UAV

In late 2011, KARI unveiled its tiltrotor Unmanned Aerial Vehicle (UAV) that can fly at around 400 km/h.

High Altitude Long Endurance (HALE) Solar-powered Unmanned Aerial Vehicle (UAV)

KARI is developing an electric-driven HALE UAV in order to secure system and operational technologies since 2010.

Optionally Piloted Personal Air Vehicle (OPPAV)

KARI leads the research and development program to develop a one-seat class electric VTOL (eVTOL) demonstrator. Flight tests of a sub-scale technology demonstrator will be started by 2021.

Future missions

Launch schedule

FY 2025

CAS500-2

CAS500-3

CAS500-4

KSLV-II-4

FY 2026

KSLV-II-5

FY 2027

COMS-3

KSLV-II-6

FY 2028

FY 2029

COMS-4

FY 2030

KSLV-III-1

FY 2032

Korean Lunar Exploration Program

Gallery

See also

Naro Space Center

Korea Astronomy and Space Science Institute (KASI)

Korea AeroSpace Administration (KASA)

Agency for Defense Development (ADD)

South Korean space program

Korean Lunar Exploration Program

List of government space agencies

References

External links

KARI official website (Korean / English)

South Korean space projects from Encyclopedia Astronautica